

ViREL-BUDGET: Reliability-Constrained Visual-Token Budgeting for Green Vision-Language Inference

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Motivation and Research Questions

Central problem. Fewer visual tokens do not automatically imply reliable or green VLM inference. Average accuracy can conceal weakened visual reliance, while pruning overhead can erase device-level energy savings.

Research questions

- Can a frozen controller choose one pruned or dense action for an unseen query while satisfying a prospective unsafe-acceptance limit?
- How do reliability-resource trade-offs vary across model scales and pruning methods?
- Which reliability-qualified configurations reproduce measured energy savings across independent workload draws?

Contributions

- Intervention-defined labels separate answer correctness from preservation of dense visual behavior.
- Budget-specific predictors select among 64, 128, 256, 432, and dense visual-token actions without VLM fine-tuning.
- Paired GPU telemetry and replication determine whether token sparsity becomes a defensible green outcome.

Why common efficiency proxies are insufficient.

Reported proxy	Unresolved deployment question
Average accuracy	Did compression preserve image-dependent behavior for this query?
Retained tokens or FLOPs	Did end-to-end latency, memory, or device energy actually decrease?
One favorable run	Does the saving persist across independent, matched workload draws?

Study thesis. Green inference couples prospective safety, one-call accounting, and energy replication.

Evaluation Design

Datasets, models, and populations.

Component	Evaluation role
POPE, MMStar, Visual CounterFact	2,100 image-question pairs with complementary visual-reliance stress tests
Development	1,200 queries for labels, predictors, and frozen risk calibration
Prospective	900 group-isolated queries for one-action reliability evaluation
Replication	Three independent, group-disjoint draws of 210 queries (630 total)
VLM backends	LLaVA-1.5-7B and 13B
Pruning substrates	FastV, SCOPE, deterministic Random
Candidate actions	64, 128, 256, 432 tokens, or dense

Measured outputs

- Reliability: accuracy, strict safety, unsafe acceptance, dense avoidance.
- Systems: GPU energy, latency, throughput, peak memory, and backend calls.
- Environmental accounting: operational CO₂e estimated from measured energy, disclosed PUE, and electricity intensity.

Protocol safeguards

- Group-aware five-fold cross-validation keeps source-related image-question instances in the same development fold.
- Feature schema, thresholds, and dense-fallback rule are frozen before the sealed prospective evaluation.
- Replication uses matched queries, matched Dense references, and counterbalanced execution order.
- Prospective labels are revealed only after controller decisions are recorded.

Separation principle. Prospective reliability and energy replication use distinct held-out populations; measured energy is reported separately from estimated carbon.

Frozen Risk-Constrained Policy

Intervention-guided safety target. During development, each candidate budget must preserve the dense answer on the original image and the dense response trajectory under eligible counterfactual, irrelevant-image, blur, and gray interventions.

$$S_{i,m,b} = A_{i,m,b} R_{i,m,b}, \quad \mathcal{B} = \{64, 128, 256, 432, \text{dense}\}.$$

Budget-specific predictors estimate

$$\hat{p}_{i,m,b} = P(S_{i,m,b} = 1 \mid x_i)$$

using only information available before VLM inference.

Controller inputs	Explicitly excluded at deployment
Question length; option count; multiple-choice flag	Dense answers and dense intervention trajectories
Dataset/task indicators; predeclared semantic keywords	Gold labels and outcomes from unexecuted budgets
Four independent L_2 logistic risk models per controller	Image-derived metadata and online post-hoc repair
Calibration rule	

$$U_W(\text{unsafe}; \tau_m) \leq \epsilon, \quad \epsilon = 0.05,$$

where U_W is the one-sided 95% Wilson upper bound. The policy selects the smallest admissible budget above the calibrated threshold; otherwise it falls back to dense inference.

One-call deployment contract. The frozen controller commits to one action before inference. It does not probe multiple budgets, use intervention outcomes online, or repair an unsafe result retrospectively.

Frozen decision sequence

1. score all four pruned budgets;
2. retain actions above τ_m ;
3. execute the smallest qualifier;
4. use Dense if none qualifies.

Strong-green gate (Gate R)

- reliability limit satisfied;
- saving positive in every draw;
- pooled 95% lower bound > 0 .

Prospective scope. The 5% gate provides empirical risk evidence under the evaluated distribution; it is not a distribution-free guarantee under arbitrary deployment shift.

Conclusions, Limitations, and Open Artifacts

Conclusions

- Visual-token budgeting should be treated as a reliability-constrained decision problem, not a universal compression ratio.
- SCOPE defines the replicated reliability-qualified energy frontier at both evaluated LLaVA scales.
- Dense avoidance and token reduction are algorithmic proxies, not green claims; device energy must be measured.

Method-specific interpretation.

SCOPE Replicated reliability-qualified energy savings at both evaluated scales.

FastV High dense avoidance without an established pooled energy reduction.

Random Uninformed control showing that token removal can increase energy.

Scope and limitations. Intervention-defined safety measures behavioral fidelity, not semantic grounding. Results remain specific to the evaluated VLMs, pruning implementations, workloads, and hardware. GPU energy is measured within declared call windows; carbon is estimated. Lifecycle impacts and cross-backbone transfer remain open.

Future validation. Richer pre-inference features and stronger risk controls should be evaluated with their extraction and probing costs inside the operational boundary, including tests under distribution shift and across VLM backbones.



GitHub code



Hugging Face data

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Selected references. [1] Chen et al., FastV, ECCV 2024. [2] Deng et al., SCOPE, NeurIPS 2025. [3] Schwartz et al., "Green AI," CACM 2020. [4] Henderson et al., JMLR 2020.

ViRel-Budget Framework

Five-layer evidence chain.

Layer	Role
Evidence	Dense original and eligible intervention trajectories define development supervision.
Actions	FastV, SCOPE, Random, and Dense share one budget set.
Predictor	Four budget-specific L_2 logistic models estimate strict safety.
Policy	The smallest qualified action is executed; Dense is the fallback.
Accounting	Reliability, latency, memory, GPU energy, carbon, and cost remain separate outputs.

Behavioral target. Strict safety preserves the dense original answer and every eligible dense intervention response; task correctness is reported separately.

Deployment semantics

- exactly one selected VLM action is executed;
- dense answers, intervention outputs, and gold labels are unavailable;
- unsafe acceptances remain visible and are never repaired retrospectively.

Risk-cost objective

$$\min_{\pi} \mathbb{E}[\pi(x)] \quad \text{s.t.} \quad P(\text{unsafe acceptance}) \leq 0.05.$$

Prospective Reliability and Coverage

$n = 900$

sealed, group-isolated prospective queries

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frozen controllers satisfy the 5% risk gate

1.67–3.44%

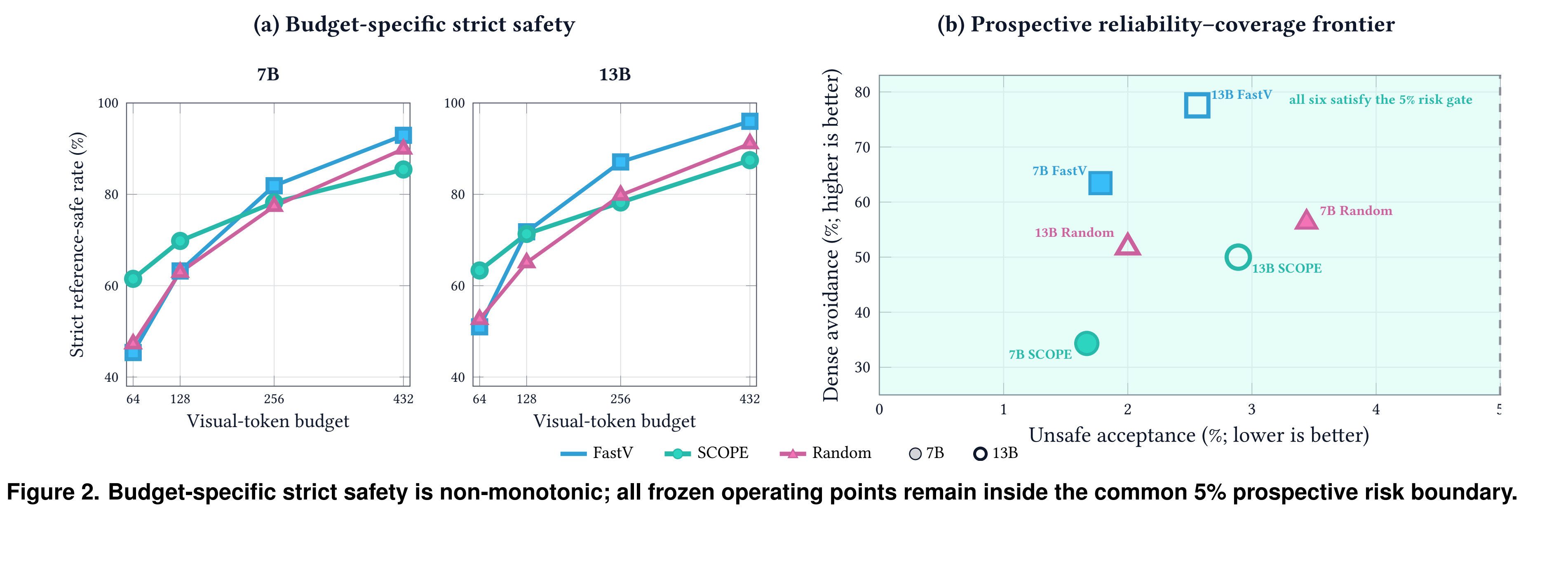
prospective unsafe-acceptance range

34.33–77.56%

reliability-qualified dense avoidance

Table 1. Frozen controllers: prospective reliability and replicated energy saving (positive means lower energy).

Model	Method	Prospective population ($n = 900$)			Replication ($n = 630$)	
		Accuracy (%)	Unsafe (%)	Dense avoid (%)	Energy saving (%)	
7B	FastV	63.67	1.78	63.44	–3.11	
	SCOPE	64.11	1.67	34.33	+11.87	
	Random	62.67	3.44	56.33	–5.96	
13B	FastV	72.56	2.56	77.56	–1.35	
	SCOPE	72.22	2.89	50.00	+10.62	
	Random	72.33	2.00	51.78	–3.92	



Replicated Green Systems Outcome

↓11.87%

7B SCOPE GPU energy
95% CI: 7.19–16.30%

↓10.62%

13B SCOPE GPU energy
95% CI: 7.48–13.46%

66 runs / 4.8198 kWh

recoverable experimental ledger; carbon reported separately

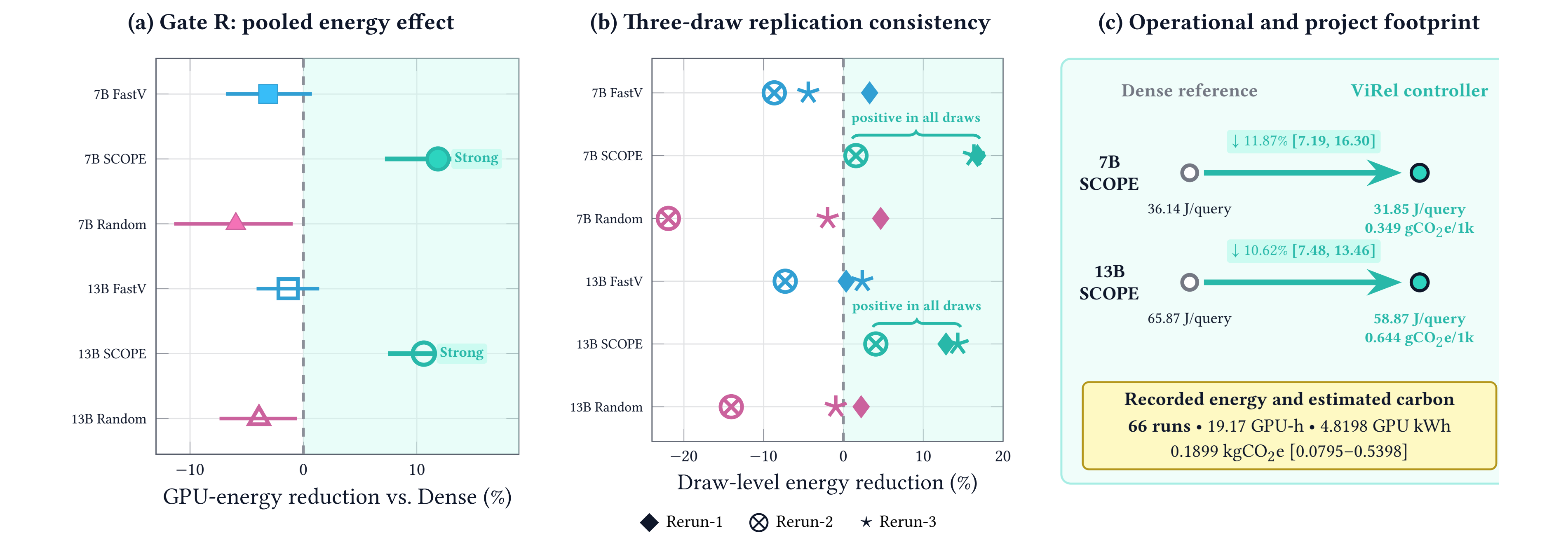


Figure 3. SCOPE consistently saves energy at both scales; project-level cost is reported separately.

Gate R adjudication. SCOPE is the only evaluated substrate with positive savings in every draw and a pooled lower confidence bound above zero at both scales. FastV and Random demonstrate that dense avoidance and token sparsity do not, by themselves, establish lower device energy.

Reliability-Qualified Pareto Frontier and Deployment Profiles

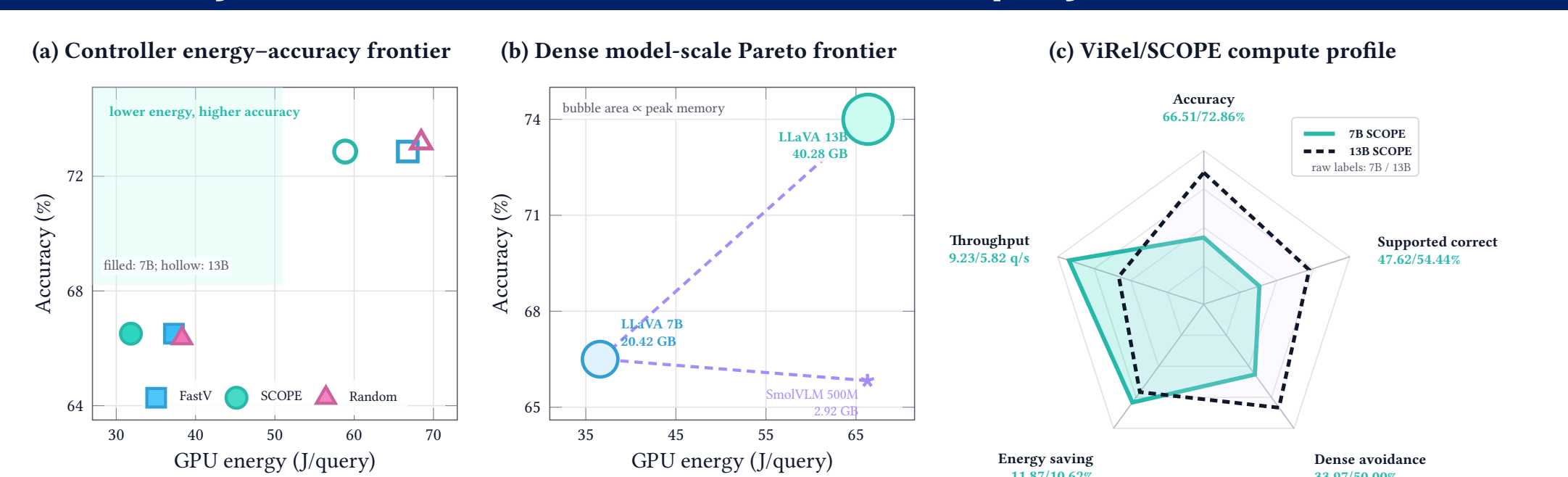


Figure 4. Reliability-qualified energy-accuracy and model-scale frontiers.

Qualification precedes comparison. Only reliability-qualified profiles enter the deployment frontier. The 7B profile favors energy productivity and throughput, whereas 13B favors accuracy, supported-correct performance, and dense avoidance; SmolVLM-500M minimizes dense peak memory. No point dominates accuracy, reliability, latency, memory, energy, and capacity simultaneously, so selection must follow a declared service objective rather than a single global efficiency score.

Validated SCOPE deployment profiles ($n = 630$).

Metric	7B	13B
Accuracy (%)	66.51	72.86
Supported-correct (%)	47.62	54.44
Dense avoidance (%)	33.97	50.00
GPU energy (J/query)	31.85	58.87
Latency (ms/query)	108.38	171.81
Throughput (queries/s)	9.23	5.82
Supported-correct/kWh	53,820	33,291